



VEYORA ACADEMY

FULL-STACK WEB & APP DEVELOPMENT

6-MONTH PROGRAM

HTML • CSS • JavaScript • React

Node.js • Express • PostgreSQL • Git



```
> build_full_stack()
```

```
<frontend />
```

```
api.listen(3000)
```

```
database.connect()
```

Build websites. Build applications. Build production-ready systems.

PROGRAM AT A GLANCE

From your first webpage to a full-stack application.

A practical 24-week developer pathway covering web foundations, JavaScript, React, backend development, databases, authentication, deployment and a complete capstone.

06 MONTHS Structured pathway	24 WEEKS Hands-on learning	07+ PROJECTS Portfolio evidence	01 CAPSTONE Full-stack build
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THE DEVELOPMENT JOURNEY

1 WEB	2 JAVASCRIPT	3 REACT	4 BACKEND	5 DATABASE	6 FULL STACK
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WHAT YOU WILL LEARN

✓ Build semantic, responsive websites with HTML and CSS.	✓ Write modern JavaScript and interactive browser applications.	✓ Build component-based interfaces with React.
✓ Create backend services and REST APIs with Node.js and Express.	✓ Design relational databases using SQL and PostgreSQL.	✓ Implement authentication, authorization and validation.
✓ Connect frontend, API and database into one full-stack system.	✓ Use Git/GitHub and deploy applications.	✓ Document and present professional portfolio projects.

WEEK 1 — HOW THE WEB WORKS	• Internet & Web basics • Client vs server • Browser, DNS and domains • HTTP / HTTPS • URLs • Requests and responses • Status codes
OUTPUT	Mini Project: Personal Profile Webpage
WEEK 2 — HTML5	• Semantic HTML • Headings, links and media • Lists and tables • Forms and input types • Accessibility basics • SEO-friendly HTML
OUTPUT	Project: Professional Profile Website
WEEK 3 — CSS	• Selectors and box model • Typography, colors and spacing • Flexbox • CSS Grid • Positioning • Responsive design • Media queries
OUTPUT	Practice: Responsive component set
WEEK 4 — RESPONSIVE WEBSITE	• Navigation • Hero section • Services • About • Contact • Footer • Mobile / tablet / desktop layouts
OUTPUT	Project: Responsive Business Website

WEEK 5 — JAVASCRIPT FUNDAMENTALS	• Variables and data types • Operators • Conditions • Loops • Functions • Arrays • Objects
OUTPUT	Practice: Logic and data exercises
WEEK 6 — MODERN JAVASCRIPT	• Array methods • Destructuring • Spread/rest • Template literals • Modules • JSON • Error handling
OUTPUT	Practice: Data transformation mini-app
WEEK 7 — DOM & BROWSER APIs	• DOM manipulation • Events • Forms and validation • LocalStorage • Fetch API • Async/await • API responses
OUTPUT	Practice: Dynamic browser application
WEEK 8 — JAVASCRIPT PROJECT	• Interactive UI • Browser APIs • API integration • Validation • Local data storage
OUTPUT	Project: To-do / Expense Tracker / Weather App

WEEK 9 — REACT FUNDAMENTALS	• Components • JSX • Props • State • Events • Conditional rendering • Lists
OUTPUT	Practice: Component-based interface
WEEK 10 — REACT APPLICATIONS	• Component architecture • Forms • Reusable components • Hooks • useState • useEffect
OUTPUT	Project: Interactive React interface
WEEK 11 — ROUTING & APIS	• React Router • Pages and navigation • API requests • Loading/error states • Environment variables
OUTPUT	Practice: Multi-page API-powered app
WEEK 12 — REACT PROJECT	• Responsive UI • Reusable components • Routing • Forms • API integration • Dynamic data
OUTPUT	Project: Professional React Web App

WEEK 13 — NODE.JS	• Node.js runtime • npm • Packages • Modules • Environment variables • Server basics
OUTPUT	Practice: Build a simple server
WEEK 14 — EXPRESS & REST APIs	• Express • Routes • Controllers • Middleware • Request/response • REST architecture • HTTP methods • Error handling
OUTPUT	Project: REST API foundation
WEEK 15 — DATABASE	• Relational databases • Tables • Primary/foreign keys • Relationships • CRUD • SQL • PostgreSQL • Database connection
OUTPUT	Practice: Build and query an application database
WEEK 16 — BACKEND PROJECT	• CRUD endpoints • Validation • Database operations • Error handling • API testing
OUTPUT	Project: Student / Product / Booking Management API

WEEK 17 — FRONTEND + BACKEND	• React → API → Node/Express → PostgreSQL • CRUD frontend • Forms • Loading/error states • Environment variables
OUTPUT	Practice: Connect React to backend
WEEK 18 — AUTHENTICATION	• Registration • Login • Password hashing concepts • Tokens / sessions • JWT concepts • Protected routes • User roles
OUTPUT	Project: Authentication flow
WEEK 19 — PRODUCTION FEATURES	• File/image upload • Search • Filtering • Pagination • Notifications • Validation • CORS • Security basics
OUTPUT	Practice: Add production-style features
WEEK 20 — FULL-STACK PROJECT	• Admin/user workflow • Dashboard • CRUD • Authentication • Database • API • Responsive frontend
OUTPUT	Project: Hotel / Business Management System

WEEK 21 — GIT & GITHUB	• Repositories • Commits • Branches • Merge • Pull requests • GitHub README
OUTPUT	Project: Publish a professional repository
WEEK 22 — DEPLOYMENT	• Production builds • Environment variables • Frontend deployment • Backend deployment • Database hosting • Domain connection • Troubleshooting
OUTPUT	Project: Deploy a working application
WEEK 23 — PROFESSIONAL DEVELOPMENT	• Code organization • Debugging • Testing basics • Performance • Security • Accessibility • SEO • API documentation • Portfolio
OUTPUT	Task: Prepare developer portfolio
WEEK 24 — FINAL CAPSTONE	• Idea → UI → Frontend → Backend → API → Database • Authentication • Testing • Deployment • Documentation • Presentation
OUTPUT	Final Project: Complete Full-Stack Application

PROJECT PORTFOLIO

Build evidence of your development skills.

01	PERSONAL PROFILE WEBSITE HTML + CSS	Semantic structure, styling and responsive layout.
02	RESPONSIVE BUSINESS WEBSITE HTML + CSS	Professional multi-section responsive website.
03	JAVASCRIPT APPLICATION JavaScript	Interactive UI + browser APIs + data.
04	REACT WEB APPLICATION React	Components + routing + API integration.
05	REST API + DATABASE Node + Express + PostgreSQL	CRUD + relational data + backend logic.
06	FULL-STACK BUSINESS APP React + Node + PostgreSQL	Frontend + API + database + authentication.
07	FINAL CAPSTONE Full Stack	Complete application + deployment.

TECH STACK

Frontend	HTML5 • CSS3 • JavaScript • React • React Router
Backend	Node.js • Express.js • REST APIs
Database	SQL • PostgreSQL
Developer Tools	VS Code • npm • Git • GitHub
Production	Environment variables • Hosting • Deployment • Domains
Professional	Authentication • Validation • Testing • Security • Accessibility • SEO

CAREER-READY OUTCOME

What you should be able to build by the end

Frontend Developer	Responsive websites and component-based React interfaces.
Backend Developer	REST APIs, server logic and database-backed applications.
Full-Stack Developer	Complete applications connecting UI, API and database.
Junior Web Developer	Git/GitHub workflow, debugging and deployment basics.
Freelance Developer	Client-ready websites and business applications with documented projects.

LEARNING METHOD

Learn → Practice → Build → Deploy → Present

Every major technology should result in a visible student output. The goal is not to memorize syntax; it is to understand how a real application is structured and how its parts work together.

VEYORA ACADEMY

A Division of VEYORA

Practical technology education focused on real skills, projects and career-ready outcomes.

Registered Business • NTN / FBR Registered

Build the web. Build the product. Build your career.